

# Saithan Panchaharan

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## EDUCATION

### University of Toronto

Sep. 2024 – June 2029 (*Expected*)

*BSc in Computer Science (Co-Op)*

*Toronto, ON*

- **Coursework:** *Shell Programming, Data Structures & Algorithms, Microprogramming, Assembly Programming, Operating Systems, OOP, Statistics & Data Analysis*

## PROJECTS

### Sportfolio · *Live Demo*

August 2026

- Built a **Spring Boot** API that models professional athletes as tradable stocks, with a pricing engine that converts real MLB performance data into daily price movements
- Designed a **PostgreSQL** database to support trading, portfolio valuation, and historical price tracking across hundreds of players
- Integrated two sports-data APIs to ingest live stats and season projections, using a caching layer to recalculate prices at no additional cost without re-fetching on every run
- Deployed the API on **Railway** and resolved production issues (server crashes from large-scale recalculations, data errors from mid-season trades) through log and API-level debugging

### LineTracker · *Live Demo*

July 2026

- Built a full-stack real-time alert platform that monitors live sports betting odds and stock prices, notifying users via email when user-defined thresholds are crossed
- Engineered a **Flask REST API** with **APScheduler** polling live data every minute across **SharpAPI** and **Yahoo Finance**, with in-memory caching to stay within rate limits and a **Supabase** backend for persistent alert storage
- Designed a multi-step betting interface that fetches, groups, and deduplicates live odds across multiple sportsbooks in real time, normalizing inconsistent team names and logos across data sources
- Implemented lightweight email-based user isolation via **localStorage** and server-side filtering, enabling multi-user support without authentication overhead

### YouTube Video Summarizer · *Live Demo*

May 2026 – June 2026

- Built a full-stack web app using **React & Flask** that fetches YouTube transcripts and generates summaries using **Claude API**
- Integrated **Supadata API** to fetch transcripts, bypassing cloud IP restrictions on server-side requests
- Deployed frontend on **Vercel** & backend on **Render** with environment-based API key management and CORS configuration for cross-origin communication
- Incorporated **Web Speech API** for text-to-speech playback & implemented transcript caching to eliminate redundant API calls when switching expertise levels

## TECHNICAL SKILLS

**Languages & Frameworks:** Python, Java, JavaScript, C/C++, SQL, Spring Boot, Flask, React.js, NumPy, Pandas

**Tools & Platforms:** PostgreSQL, MySQL, Docker, Git/GitHub, Railway, Vercel, Render, Linux

## WORK EXPERIENCE

### Digital Marketing Intern

July 2026 – Aug. 2026

*Venture for Canada*

- Developed LinkedIn content strategy and post calendars for two early-stage consumer brands, establishing a recurring content database for future scheduling
- Built **HubSpot** email marketing campaigns across two brands, including subject lines, CTAs, and click-through creative
- Designed **15+** branded ad and poster creatives (product, seasonal, and recipe-focused) aligned to each brand's style guide & marketing goals